

A continuous uniform-BV kernel whose Mercer singular-value expansion is not absolutely convergent

Candidate full counterexample to the belief stated in arXiv:2409.16453

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Status. Candidate full counterexample, likely valid. There is a real continuous kernel on $[-\pi, \pi]^2$ whose sections have uniformly bounded variation and whose Mercer singular-value expansion converges uniformly, but not absolutely at $(\pi/4, \pi/4)$. Its positive singular values are simple, so the failure cannot be removed by choosing a different singular basis.

1 The source statement

Jeong and Townsend consider a continuous kernel $K : [a, b] \times [c, d] \rightarrow \mathbb{R}$ of *uniform bounded variation*, meaning that its weak partial derivatives satisfy

$$\sup_y \int_a^b |\partial_x K(x, y)| dx < \infty, \quad \sup_x \int_c^d |\partial_y K(x, y)| dy < \infty.$$

They prove almost-everywhere and almost-uniform convergence results and then state their belief that the Mercer singular-value expansion of every such kernel converges “pointwise, absolutely, and uniformly” [1, p. 2]. The complete context, including their definition, is reproduced in Figure 1.

2 Proof intuition

Absolute SVE convergence would force the values of the singular functions to compensate for a borderline nonsummable singular spectrum. Trigonometric singular functions allow us to defeat that compensation at one explicit point. We choose singular coefficients

$$a_n = \frac{1}{n \log(n+2)},$$

which are square-summable but not summable on the odd integers. The resulting sine series nevertheless converges uniformly because $na_n \rightarrow 0$.

The only subtlety is bounded variation: formally differentiating loses the factor $1/n$ and produces coefficients $b_n = 1/\log(n+2)$, which are not summable. Convexity of (b_n) resolves this. A positive mixture of the Fejér kernels has exactly these Fourier coefficients and finite L^1 norm. Its mean-zero part is therefore the weak derivative of a continuous periodic primitive f . The identity

$$K(x, y) = \frac{f(x+y) + f(x-y)}{2}$$

then supplies the uniform variation bounds, while orthogonality identifies the series as the actual SVE. At $(\pi/4, \pi/4)$ its absolute values contain one half of the divergent odd subseries of $\sum a_n$.

Surprisingly, a continuous symmetric kernel may not have a Mercer-like expansion that converges pointwise (see Proposition 1). We hope this new example clarifies some confusion in the literature on Mercer’s expansion for general kernels. We also prove that general continuous kernels may also have Mercer’s expansions that converge pointwise, but not uniformly and absolutely, meaning that Mercer’s expansion must be used with some care for general kernels.

This work provides a rigorous theoretical underpinning of Mercer’s expansion for indefinite and asymmetric kernels. For indefinite kernels, the eigenfunction expansion is the most natural generalization of Mercer’s expansion. For asymmetric kernels, the most natural generalization is the following singular value expansion (SVE) that was considered in the early 1900s (Schmidt, 1907)

$$K(x, y) = \sum_{n=1}^{\infty} \sigma_n u_n(x) v_n(y), \quad (2)$$

where $\sigma_1 \geq \sigma_2 \geq \dots > 0$ are the singular values of K and $\{u_n\}_{n \in \mathbb{N}}$ and $\{v_n\}_{n \in \mathbb{N}}$ are sets of orthonormal functions called the right and left singular functions of K , respectively. The expansion in eq. (2) can replace Mercer’s expansion for asymmetric kernels (see Section 3). We regard eq. (2) as a natural generalization of Mercer’s expansion to general continuous kernels.

Schmidt proved the existence of the expansion in eq. (2) for all square-integrable kernels and showed that the equality holds in the L^2 sense (Schmidt, 1907). He also proved that for continuous kernels K , if $\sigma_n > 0$, the corresponding left and right singular functions can be selected as continuous. Later, in 1923, Hammerstein showed that the expansion in eq. (2) converges uniformly for any continuous kernel that also satisfies a contrived square-integrated Lipschitz-like condition (Hammerstein, 1923). Later, in 1938, Smithies gave a finicky condition on continuous kernel that ensures specific decay rates of the singular values, and he used that decay to prove almost everywhere pointwise and almost everywhere absolute convergence (Smithies, 1938). More recently, it was realized that both Hammerstein’s and Smithies’ conditions are satisfied by continuous kernels that are Lipschitz continuous uniformly in each variable separately (Townsend, 2014).

We prove results for a larger class of general continuous kernels $K : [a, b] \times [c, d] \rightarrow \mathbb{R}$ than Lipschitz continuous kernels. We only need the kernels to have bounded variation uniformly in each variable separately, i.e., for any fixed $x \in [a, b]$ and any fixed $y \in [c, d]$, we have

$$\int_a^b \left| \frac{\partial}{\partial x} K(x, y) \right| dx < V, \quad \int_c^d \left| \frac{\partial}{\partial y} K(x, y) \right| dy < V, \quad (3)$$

for some fixed uniform constant $V < \infty$. We say a kernel is of uniform bounded variation if it satisfies the conditions in eq. (3). The partial derivatives in eq. (3) are replaced by the notion of weak derivatives if the kernel is not differentiable (Trefethen, 2019, Chap. 7). We also provide new decay bounds on σ_n for smoother kernels (see Section 4), which we believe are asymptotically tight, and can be used to rigorously truncate Mercer’s expansion.

We prove a new result that continuous kernels of uniform bounded variation have a Mercer’s expansion that converges pointwise almost everywhere, unconditionally almost everywhere, and almost uniformly. We have not been able to prove that the expansion converges pointwise, absolutely, and uniformly, so there could still be some subtleties in using Mercer’s expansion for general kernels. However, we believe that continuous kernels of uniform bounded variation have a Mercer’s expansion that converges pointwise, absolutely, and uniformly, and we hope later research will prove this.

1.1 OUR CONTRIBUTIONS

We generalize Mercer’s expansion to a general continuous kernel, where the kernel may be indefinite or asymmetric. The following two theorems summarize our main theoretical results about Mercer’s expansion for general continuous kernels (see Section 3).

Theorem. *For any $[a, b] \subset \mathbb{R}$ there are continuous symmetric indefinite kernels on $[a, b] \times [a, b]$ such that Mercer’s expansion in eq. (2): (i) does not converge pointwise, (ii) converges pointwise but not uniformly, and (iii) converges pointwise but not absolutely.*

Theorem. *For any $[a, b], [c, d] \subset \mathbb{R}$, let $K : [a, b] \times [c, d] \rightarrow \mathbb{R}$ be a continuous kernel of uniform bounded variation (see eq. (3)). Then, Mercer’s expansion of K in eq. (2) converges pointwise, unconditionally almost everywhere, and almost uniformly.*

Figure 1: Full-width rendering of PDF page 2 of arXiv:2409.16453v2. The uniform-BV definition is equation (3), and the stated belief appears in the paragraph immediately below it.

3 The counterexample

Theorem 1. *There is a real continuous asymmetric kernel $K : [-\pi, \pi]^2 \rightarrow \mathbb{R}$ of uniform bounded variation such that its Mercer singular-value expansion converges uniformly on the square but is not absolutely convergent at $(\pi/4, \pi/4)$. Its nonzero singular values are all simple.*

We prove the theorem in four steps.

3.1 An integrable function with prescribed slow Fourier coefficients

Put

$$b_n = \frac{1}{\log(n+2)} \quad (n \geq 0), \quad \Delta^2 b_N = b_N - 2b_{N+1} + b_{N+2}.$$

The function $x \mapsto 1/\log(x+2)$ is strictly decreasing and

$$\frac{d^2}{dx^2} \frac{1}{\log(x+2)} = \frac{\log(x+2) + 2}{(x+2)^2 \log^3(x+2)} > 0.$$

Thus $b_n \downarrow 0$ and $\Delta^2 b_N > 0$. Let

$$F_N(t) = 1 + 2 \sum_{k=1}^N \left(1 - \frac{k}{N+1}\right) \cos(kt) = \frac{1}{N+1} \left(\frac{\sin((N+1)t/2)}{\sin(t/2)} \right)^2$$

with the continuous interpretation at multiples of 2π . These are the nonnegative Fejér kernels [2], normalized by $\int_{-\pi}^{\pi} F_N = 2\pi$. Define, in $L^1(\mathbb{T})$,

$$g(t) = \frac{1}{2} \sum_{N=0}^{\infty} (N+1) \Delta^2 b_N F_N(t). \quad (1)$$

To justify convergence and compute the coefficients, write $d_N = b_N - b_{N+1}$. For finite M ,

$$\sum_{N=0}^M (N+1) \Delta^2 b_N = b_0 - b_{M+1} - (M+1)d_{M+1}, \quad (2)$$

$$\sum_{N=n}^M (N+1-n) \Delta^2 b_N = b_n - b_{M+1} - (M+1-n)d_{M+1}. \quad (3)$$

The mean-value theorem gives $Nd_N \rightarrow 0$. Hence the nonnegative terms in (1) have total L^1 norm πb_0 , and the series converges in L^1 . Equations (2)–(3) show that g has constant term $b_0/2$ and cosine coefficient b_n for every $n \geq 1$. Consequently

$$h = g - \frac{b_0}{2} \quad (4)$$

is even, belongs to $L^1(\mathbb{T})$, has mean zero, and has cosine coefficients b_n .

3.2 A continuous periodic primitive and its uniform series

Define $f(t) = \int_0^t h(s) ds$ on $[-\pi, \pi]$ and extend it periodically. Evenness and zero mean of h give $f(-t) = -f(t)$ and $f(-\pi) = f(\pi) = 0$. Thus f is continuous, periodic, and absolutely continuous, with weak derivative h . Integration by parts shows that its n th sine coefficient is

$$a_n = \frac{b_n}{n} = \frac{1}{n \log(n+2)}. \quad (5)$$

For completeness, we verify uniform convergence rather than cite a criterion. If (a_n) is positive decreasing and $A_N = \sup_{n \geq N} n a_n \rightarrow 0$, then for $0 < t \leq \pi/2$ split a tail at $q = \lfloor 1/t \rfloor$. On $N \leq n \leq q$, the bound $|\sin(nt)| \leq nt$ gives a total at most A_N . On the remaining indices, Abel summation and

$$\left| \sum_{n=r}^s \sin(nt) \right| \leq \frac{2}{|\sin(t/2)|}$$

give at most a fixed multiple of A_N . If $t > 1/N$, the same Abel estimate applied from N gives the bound directly. For $\pi/2 \leq t \leq \pi$, the partial sine sums are uniformly bounded, so Abel summation again applies. The endpoints are trivial. Therefore

$$\sup_{t \in \mathbb{R}} \left| \sum_{n=N}^M a_n \sin(nt) \right| \leq C A_N \quad (M \geq N) \quad (6)$$

for an absolute C .

Here $n a_n = b_n \rightarrow 0$, so $\sum_{n \geq 1} a_n \sin(nt)$ converges uniformly to a continuous periodic function S . Both S and f are odd and have the same sine coefficients. Their difference has every Fourier coefficient zero. Convoluting that continuous difference with $F_N/(2\pi)$ gives zero for every N , while the displayed positive-kernel formula makes these convolutions converge uniformly to the difference. Hence

$$f(t) = \sum_{n=1}^{\infty} a_n \sin(nt) \quad (7)$$

uniformly on $\mathbb{R}$.

3.3 The kernel and its uniform variation bounds

Set

$$K(x, y) = \frac{f(x+y) + f(x-y)}{2} = \sum_{n=1}^{\infty} a_n \sin(nx) \cos(ny). \quad (8)$$

The identity and uniform convergence follow from (7). In particular K is real and continuous. Its weak partial derivatives are

$$\partial_x K(x, y) = \frac{h(x+y) + h(x-y)}{2}, \quad \partial_y K(x, y) = \frac{h(x+y) - h(x-y)}{2}.$$

Because an interval of length 2π contains one full period of every translate of h ,

$$\int_{-\pi}^{\pi} |\partial_x K(x, y)| dx \leq \|h\|_{L^1(\mathbb{T})}, \quad \int_{-\pi}^{\pi} |\partial_y K(x, y)| dy \leq \|h\|_{L^1(\mathbb{T})}.$$

The bounds are uniform in the fixed variable, exactly as required by the source definition. The kernel is asymmetric: $K(0, y) = 0$ for every y , while $K(y, 0) = f(y)$ is not identically zero because its first sine coefficient is $a_1 > 0$.

3.4 Exact SVE and failure of absolute convergence

Let $T_K : L^2[-\pi, \pi] \rightarrow L^2[-\pi, \pi]$ be $(T_K \varphi)(x) = \int_{-\pi}^{\pi} K(x, y) \varphi(y) dy$, and put

$$u_n(x) = \frac{\sin(nx)}{\sqrt{\pi}}, \quad v_n(y) = \frac{\cos(ny)}{\sqrt{\pi}}.$$

Orthogonality and (8) give

$$T_K v_n = \pi a_n u_n, \quad T_K^* u_n = \pi a_n v_n. \quad (9)$$

Moreover, T_K is Hilbert–Schmidt, vanishes on the constant and sine subspaces, and its range is contained in the sine subspace. Since the constant, sine, and cosine systems give the standard orthogonal decomposition of $L^2[-\pi, \pi]$, the complete list of positive singular values is

$$\sigma_n = \pi a_n,$$

and the SVE term $\sigma_n u_n(x) v_n(y)$ is exactly $a_n \sin(nx) \cos(ny)$. The sequence (a_n) is strictly decreasing, so each positive singular value is simple; only simultaneous sign changes of u_n, v_n are possible.

Finally, at $x = y = \pi/4$,

$$\begin{aligned} \sum_{n=1}^{\infty} |\sigma_n u_n(x) v_n(y)| &= \sum_{n=1}^{\infty} a_n |\sin(n\pi/4) \cos(n\pi/4)| \\ &= \frac{1}{2} \sum_{\substack{n \geq 1 \\ n \text{ odd}}} \frac{1}{n \log(n+2)} = \infty. \end{aligned}$$

For instance, after including the prefactor $1/2$, the $n = 2k+1$ summand dominates $1/[8(k+1) \log(k+2)]$, whose sum diverges. This proves Theorem 1.

4 Verification, scope, and novelty

Eight focused audits checked independently: positivity and L^1 convergence of (1); every boundary term in (2)–(3); the uniform tail estimate; identification of the primitive; both weak-variation bounds; the exhaustive SVD; the simple-spectrum divergent point; and periodic endpoint/scope issues. The accompanying checker verifies high-precision finite instances of the telescoping identities, monotonicity and convexity, sampled uniform tails, simple coefficients, and growing absolute partial sums. These computations are sanity checks, not ingredients of the proof.

The counterexample lies in the paper’s general asymmetric-kernel class and therefore disproves the stated universal belief. It does not address the narrower question restricted to symmetric kernels. It is compatible with the authors’ theorem: the failure occurs at one point, while their unconditional conclusion is only almost everywhere.

Novelty is provisional. Bounded searches through 13 August 2026 covered the local run indexes and source corpus, the exact title and arXiv id, the quoted uniform-BV/absolute-convergence statement, close Fourier/SVE phrases, the arXiv and OpenReview records, and title/citation searches. No later answer or matching construction was found. Priority is not claimed. Human review should first check the coefficient identity (3) and the exhaustiveness statement following (9).

References

- [1] S. Jeong and A. Townsend, *Extending Mercer’s expansion to indefinite and asymmetric kernels*, Proceedings of ICLR 2025; arXiv:2409.16453v2.
- [2] L. Fejér, *Untersuchungen über Fouriersche Reihen*, Mathematische Annalen **58** (1904), 51–69.